

Borhan Javadian

CYBERSECURITY & CRYPTOGRAPHY RESEARCHER

Istanbul, Türkiye

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Education

Sabancı University

Istanbul, Türkiye

B.SC. IN COMPUTER SCIENCE AND ENGINEERING

Oct 2023 – Present

- GPA: 3.73 / 4.00 (as of July 2025) — consistently increasing; core CS GPA estimated at 3.95+.
- Expected Graduation: June 2027.
- Relevant Coursework: Cryptography, Network Security, Blockchain and Security Applications, Malware Analysis, Advanced Algorithms.

Sabancı University

Istanbul, Türkiye

MINOR IN MATHEMATICS

Oct 2023 – Present

- Complementary concentration focused on Number Theory, Abstract Algebra, and Probability.
- Emphasized mathematical foundations for cryptography, homomorphic encryption, and computational theory.

Research Experience

RESEARCH INTERESTS

- Fully Homomorphic Encryption (FHE); Post-Quantum Cryptography; Secure Computation; Zero-Knowledge Proofs; Blockchain Security; Privacy-Preserving Machine Learning.

Sabancı University

Istanbul, Türkiye

RESEARCH INTERN – PURE PROGRAM (PROJECT ON HOMOMORPHIC ENCRYPTION OPTIMIZATION)

Jul 2025 – Sept 2025

- Investigated computational bottlenecks in matrix-vector multiplication under the CKKS scheme.
- Designed and benchmarked chunking strategies for optimizing encrypted linear algebra.
- Implemented and tested performance gains using Microsoft SEAL and C++.
- Poster:** “Optimizing FHE Matrix Multiplication (CKKS)” presented at the PURE Research Symposium (2025).
- Included runtime benchmarks and performance analysis for encrypted computation under CKKS.

Sabancı University

Istanbul, Türkiye

RESEARCH ASSISTANT – MATRIX MULTIPLICATION FOR FHE

Sept 2025 – Present

- Conducting research on homomorphic matrix operations under the CKKS scheme using bicyclic encoding.
- Developing modular C++/SEAL framework for efficient encrypted matrix-vector multiplication.
- Investigating diagonal-based packing strategies and ciphertext rotation optimizations.

Selected Projects

ENCSUITE — GO-BASED CRYPTOGRAPHY SUITE (ENCSUITE-GO)

- Developing a modular Go library and web interface implementing classical and modern cryptographic algorithms.
- Built for education and reproducible research; includes hashing, symmetric, and asymmetric encryption demos.
- GitHub:** borhanxj/encsuite-go | **Demo:** borhanxj.github.io/encsuite-go

DEFI BLOCKCHAIN APP — SOLIDITY & REACT

- Smart contract-based lending and staking dApp on Ethereum Sepolia using Web3.js and MetaMask.
- GitHub:** borhanxj/defi-blockchain-app

CONCURRENT PORT SCANNER (GO-PORTSCANNER) — GO

- High-performance TCP scanner leveraging goroutines, worker pools, and context cancellation.
- Improved efficiency and scalability compared to earlier Python version.
- GitHub:** borhanxj/go-portscanner

STRESS LEVEL ANALYSIS — DATA SCIENCE PROJECT

- Explored correlations between stress, inflation, and social media using multi-dataset Python EDA.
- GitHub:** borhanxj/stress-level-analysis

Workshops & Conferences

enCRYPTON Project • European Union Funded

Conrad Istanbul Bosphorus, Türkiye

PARTICIPANT — LIGHTSEC 2025: 6TH INTERNATIONAL WORKSHOP ON LIGHTWEIGHT CRYPTOGRAPHY FOR SECURITY & PRIVACY

Sept 2025

- Attended a two-day international workshop gathering 80+ researchers from 18 countries on lightweight and post-quantum cryptography.
- Engaged in technical sessions on homomorphic encryption, zero-knowledge proofs, blockchain consensus, and cryptanalysis.

Sabancı University

Istanbul, Türkiye

QUANTUM OPTIMIZATION AND SECURITY CONFERENCE — DEAN'S SPEAKER SERIES

Spring 2025

- Attended keynote by Dr. Özlem Salehi Köken on quantum optimization and post-quantum cryptography research.

Awards & Honors

ACADEMIC SUCCESS SCHOLARSHIP (50%)

- Awarded by Sabancı University for outstanding GPA and consistent academic excellence (2025–Present).

MERIT-BASED SCHOLARSHIP (25%)

- Granted upon admission for exceptional academic performance (2023–Present).

DEAN'S HONOR LIST

- Multiple semesters with High Honors (GPA > 3.50).

1ST PLACE — CARIBOU MATHEMATICS CONTEST

- Achieved world top rank in High School Division (2021).

DISTINCTION — FERMAT MATHEMATICS CONTEST

- Certificate of Distinction from the University of Waterloo (2021).

Technical Skills

Programming	Go, C++, Python, Solidity, Bash, JavaScript
Cryptography	Fully Homomorphic Encryption , Post-Quantum Cryptography, Zero-Knowledge Proofs
Research & Development	Microsoft SEAL, Cryptographic Benchmarking, Algorithm Optimization, Parallel Computing (CUDA)
Software Engineering	Git, Linux, Modular Design, API Integration, Web3
Mathematics	Number Theory, Abstract Algebra, Linear Algebra, Probability, Discrete Mathematics
Data Science & Visualization	Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis

Teaching Experience

Sabancı University

Istanbul, Türkiye

LEARNING ASSISTANT — DIGITAL SYSTEMS DESIGN (CS 303)

Spring 2024 – Present

- Supported lab sessions on combinational and sequential circuit design using logic gates and Verilog.
- Guided students in debugging hardware simulations and FPGA-based implementations.
- Collaborated with instructors to develop problem sets and exam review materials.

Sabancı University

Istanbul, Türkiye

LEARNING ASSISTANT — CALCULUS I (MATH 101)

Spring 2023

- Conducted weekly recitation sessions covering limits, derivatives, and integrals.
- Held office hours and problem-solving workshops for first-year engineering students.
- Assisted with grading and provided individual tutoring to reinforce conceptual understanding.

Languages

Persian	Native
English	Full Professional
Turkish	Full Professional